

LiFi Based Automatic Traffic Signal Control For Ambulance

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Abstract—India is one of the fastest developing country in the world. One of the disadvantages of this unprecedented growth is the ineffective traffic management. Road traffic congestion becomes a major issues for highly crowded metropolitan cities like, Bengaluru. Ambulance service is one of the major services which gets affected by traffic jams. Ambulance needs to be on time as it is life deciding factors in emergency cases. In India, many lives have been lost as emergency vehicles like ambulance or firefighting vehicles were not able to reach their destination on time due to heavy traffic. An efficient and feasible traffic management system can be implemented using Li-Fi.

Light Fidelity (Li-Fi) is a form of bi-directional Visible Light Communication (VLC) in which light is modulated at speeds greater than a human eye can follow. Li-Fi can be used to implement Vehicle to Vehicle (V2V) communication as it has many advantages over other communication protocols. One of the main advantages of Li-Fi is that it provides connectivity within a very large area with more security and higher data rates. Emergency vehicles such as Ambulances, Firefighting trucks, Police vehicles etc. can propagate faster through traffic-dense roads using Li-Fi based V2V communication system.

Index Terms—Li-Fi, V2V, VLC, traffic management, emergency vehicle

I. INTRODUCTION

Wireless communication has traditionally relied on radio waves, but with advancements in technology and increasing traffic density, the radio spectrum has become congested. To manage traffic more efficiently, a reliable wireless V2V communication system is needed. However, implementing V2V communication using radio waves requires a significant amount of bandwidth, which is difficult due to the congestion caused by human cellular communication systems. Additionally, the associated circuitry for radio wave communication can be expensive. As a result, there is a growing need for alternative spectrum options. Gamma rays, X-rays, infra-red, and ultraviolet rays are not suitable for V2V communication due to their harmful effects on the human body.

Visible light communication (VLC) is an emerging platform for wireless communication that utilizes visible light frequencies ranging from 400 to 800 THz. Li-Fi is a VLC protocol

that shares many similarities with the Wi-Fi protocol, except for the physical layer. The concept of Li-Fi was introduced by Dr. Harald Haas in a 2011 TED Global speech, where he emphasized the use of LED bulbs as the primary light source due to their high luminosity and energy efficiency compared to older bulbs like incandescent and fluorescent. Li-Fi technology takes advantage of the high-speed modulation capabilities of LED bulbs to transmit data at high speeds and enable parallel transmission of thousands of data streams. The receiving element in Li-Fi technology is a photodiode, which is a cost-effective device that makes Li-Fi an economically viable solution for the challenges posed by radio wave communication.

Numerous lives have been lost in India due to the inability of emergency vehicles, such as ambulances and firefighting vehicles, to reach their intended destination on time amidst heavy traffic. To address this issue, a practical traffic management system utilizing Li-Fi technology can be implemented. This paper presents the implementation of a Li-Fi-based system for detecting emergency vehicles and facilitating their timely passage through traffic.

II. LITERATURE REVIEW

Studies on traffic management, V2V communication, and ambulance-based traffic control have been conducted since the 1990s. To implement emergency vehicle-based traffic control, different communication protocols such as GSM, RFID, IR, and ZigBee have been explored. However, these methods are no longer viable in the current world due to various reasons.

Cellular communication systems have extensively utilized GSM-based technology. However, compared to other communication protocols, the infrastructure required for GSM communication is high. While basic infrastructure for GSM is available, V2V communication utilizing this technology requires a high bandwidth [2]. In contrast, RFID-based V2V communication does not require a line of sight and can detect emergency vehicles if within the frequency range of the traffic control. However, RFID messages are vulnerable

to unauthorized access, and the metallic chassis of vehicles can cause interference with the RFID transceiver [14]. IR-based emergency vehicle detection is unreliable as it requires a line of sight and sunlight can interfere with IR signals [6]. ZigBee-based traffic management is a more cost-effective alternative that offers a wide range of communication and does not require a line of sight. Nevertheless, implementing V2V communication or traffic management using ZigBee involves restrictions on the number of nodes per controller [4].

Several research papers have suggested the potential use of Li-Fi in emergency vehicle-based traffic control systems. Unlike GSM, Li-Fi does not require any additional infrastructure and offers a higher level of security than RFID [14]. Additionally, it is more cost-effective compared to ZigBee and does not use the congested 2.4 GHz frequency band. This paper describes the implementation of a Li-Fi-based V2V communication system for intelligent traffic signal control.

III. PROBLEM IN EXISTING SYSTEM

There exist several methods to control traffic signals in the current system. These methods are as follows:

1. Manual control is a conventional and fundamental method for controlling traffic signals. It involves a human operator who manually changes the signal timing based on their judgment of the traffic flow. However, this method is susceptible to human error, which can lead to inefficiencies and delays in traffic flow.

2. Fixed-time control is a technique where the signal timing is pre-determined and fixed, regardless of the actual traffic volume. This method is relatively easy to implement and straightforward, but it does not consider variations in traffic volume throughout the day. As a result, it can cause delays and congestion during peak traffic periods and wasted capacity during off-peak hours.

3. Actuated control is a more advanced method that utilizes sensors like loop detectors or cameras to detect the presence of vehicles or pedestrians at the intersection. The signal timing is then adjusted based on real-time traffic flow, resulting in more efficient use of the intersection capacity. However, this method can be expensive to implement and maintain, and it relies on the accuracy and functionality of the sensors used.

4. Adaptive control is the most advanced method of traffic control that uses a centralized computer system to monitor real-time traffic flow and adjust signal timing accordingly. This method is more efficient than fixed-time or actuated control as it can respond quickly to changes in traffic volume and adjust the signal timing accordingly. However, it requires significant investment in infrastructure and technology to implement and maintain, and it relies on accurate real-time traffic data and advanced algorithms.

In conclusion, manual control, fixed-time control, actuated control, and adaptive control are all traffic control methods that have advantages and limitations. While manual and fixed-time control are relatively simple and cost-effective, they can result in inefficiencies and delays during peak traffic periods. Actuated control is a more advanced method that can adjust

signal timing based on real-time traffic flow, but it can be costly to implement and maintain. Adaptive control is the most advanced method that can respond quickly to changes in traffic flow, but it requires significant investment in infrastructure and technology to implement and maintain.

IV. METHODOLOGY

A. Working Principle of Li-Fi

Li-Fi (Light Fidelity) is a wireless communication technology that uses visible light to transmit data. It works by modulating the light emitted by LEDs (Light Emitting Diodes) to encode data, which is then received and decoded by a photodetector. The principle of Li-Fi is based on the concept of visible light communication (VLC), which refers to any type of wireless communication that uses visible light between 400 and 800 THz (terahertz) to transmit data.

The basic principle of Li-Fi communication is the use of binary signals, where the high and low intensity of light is used to represent the digital 1s and 0s. Li-Fi can achieve high-speed data transmission, up to several gigabits per second, which is much faster than traditional Wi-Fi technology.

The working of Li-Fi is simple. A light source, typically an LED, is modulated to encode data in the form of light pulses. The light pulses are then received by a photodetector, which converts them back into electrical signals for further processing. The receiver detects the changes in the light intensity and decodes the data from the light signals.

B. Block Diagram

The block diagram shown in Figure 1 outlines the proposed configuration of a traffic management system that employs Li-Fi technology. The system comprises two key nodes: an ambulance (an emergency vehicle) and a traffic signal control. The nodes are powered by microcontrollers, namely ArduinoUno ATmega328, which are responsible for controlling the transmission and reception of data.

The ambulance node incorporates a Li-Fi transmitter, such as an LED lamp, that receives command signals from the microcontroller. These signals carry vital information about the direction from which the ambulance is approaching the traffic signal.

In contrast, the traffic signal node is equipped with a Li-Fi receiver, for example, a solar panel, which receives signals from the ambulance's transmitter. The signals are then decrypted by the microcontroller located within the traffic signal node. Once the signal has been decoded, the microcontroller triggers the traffic signal LED light to alter the color of a particular direction to green, allowing the ambulance to pass through smoothly.

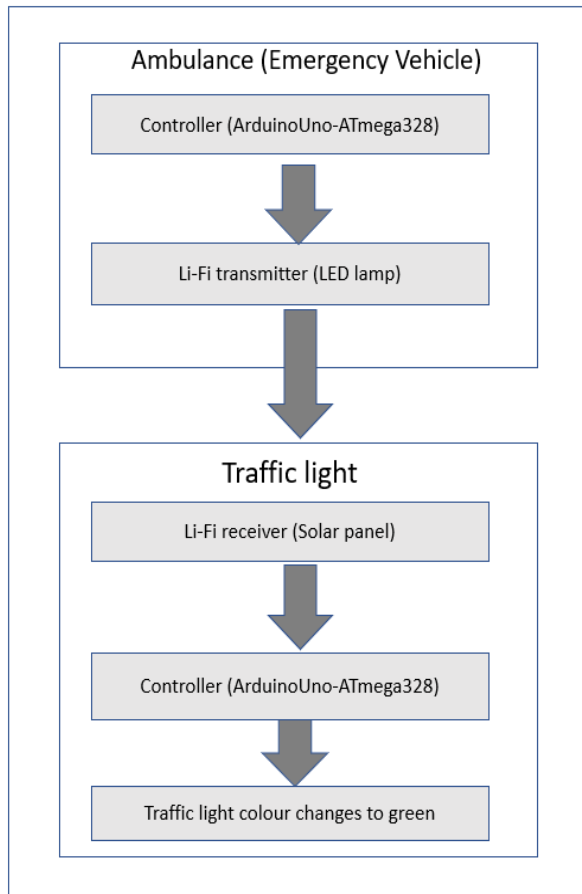


Fig. 1. Block Diagram of Li-Fi based Traffic control system

C. Components

1. Micro-controller - Arduino Uno ATmega328

The ATmega328-based Arduino Uno microcontroller board includes several components such as 14 digital input/output pins (with 6 that can function as PWM outputs), 6 analog inputs, a 16 MHz ceramic resonator, a USB connection, a power jack, an ICSP header, and a reset button. Connecting the board to a computer via USB or powering it with an AC-to-DC adapter or battery allows for immediate use. Unlike previous boards, the Uno does not use the FTDI USB-to-serial driver chip, but instead features the Atmega16U2 (Atmega8U2 up to version R2) as a USB-to-serial converter.

The Arduino Uno microcontroller board can be powered in two ways: via the USB connection or with an external power supply. The selection of the power source is automatic. An external power supply can be connected through either an AC-to-DC adapter (wall-wart) or a battery. If using an adapter, a 2.1mm center-positive plug can be inserted into the board's power jack, while battery leads can be inserted into the Gnd and Vin pin headers of the POWER connector.

The board can run on an external power supply of 6 to 20 volts, but if less than 7V is supplied, the 5V pin may output less than 5 volts and the board may be unstable. Using more than 12V can also cause the voltage regulator to overheat and

damage the board. The recommended power supply range is 7 to 12 volts.

The power pins on the board include VIN, which is the input voltage when using an external power source; 5V, which outputs a regulated 5V from the onboard regulator and can be supplied power from the DC power jack (7-12V), the USB connector (5V), or the VIN pin of the board (7-12V); 3V3, which generates a 3.3 volt supply through the onboard regulator with a maximum current draw of 50 mA; and GND, which is the ground pin. It is important to note that supplying voltage through the 5V or 3.3V pins bypasses the regulator and can potentially damage the board, which is not recommended.

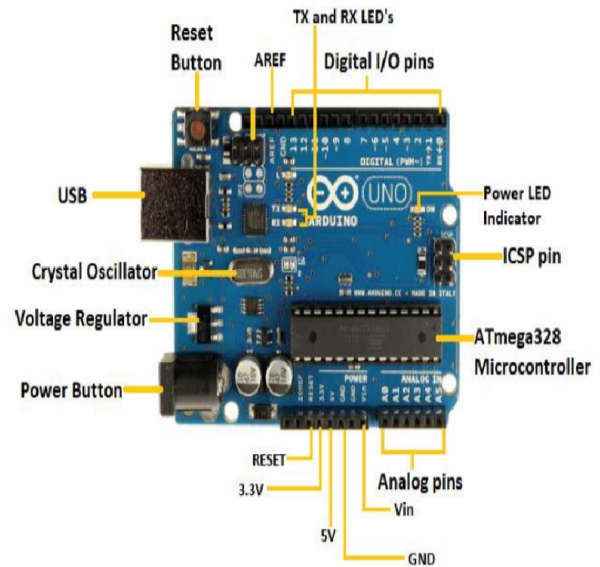


Fig. 2. Pin diagram of Arduino Uno ATmega328 Microcontroller

2. Li-Fi transmitter - LED lamp

The use of LED lamps as Li-Fi transmitters in a traffic signal control system can provide reliable and efficient communication, especially when the baud rate is set at 1200. With this baud rate, the LED lamp can transmit data at a faster rate than lower baud rates, ensuring quick and accurate transmission of information.

Moreover, the LED lamp is an affordable and low-power option for a Li-Fi transmitter, making it an ideal choice for traffic signal control systems. Additionally, it can easily be installed in existing traffic lights, simplifying the setup process.

When used in the context of an ambulance, the LED lamp can quickly communicate information about its location and direction to the traffic signal control node, enabling the traffic signal to adjust accordingly. This can help to prevent accidents and ensure a safe and efficient flow of traffic.

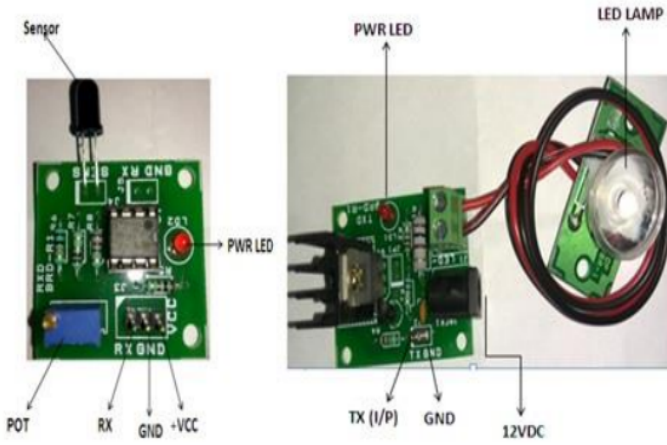


Fig. 3. LED lamp with power source

3. Li-Fi receiver - Solar Panel

In a Li-Fi based traffic management system for ambulances, a solar panel can function as a Li-Fi receiver for the traffic signal control node. As the Li-Fi transmitter in the ambulance emits modulated light signals, the solar panel receives these signals by detecting the variations in light intensity. This ability to detect and interpret light signals enables the solar panel to act as a receiver of data. The signals are then decrypted and processed by the microcontroller in the traffic signal control node to direct the traffic light accordingly. The use of a solar panel ensures a self-sufficient and sustainable power source for the traffic signal control node. This makes it an energy-efficient and cost-effective option for a reliable means of data transmission.

In order to make a Li-Fi receiver using a solar panel, one can follow the steps provided in the given circuit diagram. Firstly, solder wires to the positive (+ve) and negative (-ve) terminals of the solar panel. Next, connect the negative terminal to the ground terminal of the other jack. Finally, connect the positive terminal of the solar panel to the common wire of the left and right terminals from the jack to complete the circuit. This setup allows the solar panel to receive Li-Fi signals and convert them into electrical signals that can be further decrypted and processed by the microcontroller in the traffic signal control.

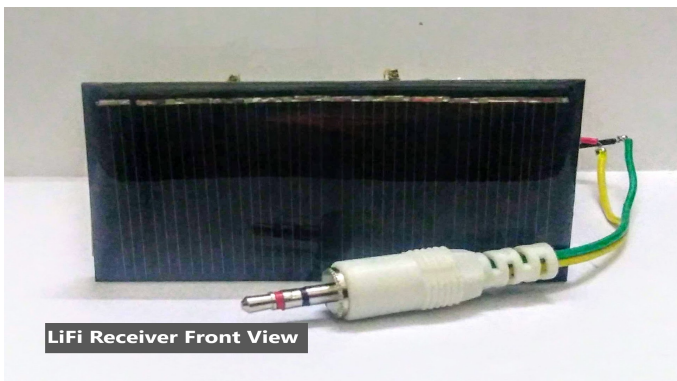


Fig. 4. Solar panel as Li-fi receiver

V. IMPLEMENTATION

1. Ambulance and Traffic light communication using Li-Fi: Let's consider a scenario where an ambulance is traveling from the south towards the north direction while the traffic signals for the lanes are switching alternatively in a regular manner. To prioritize the ambulance's movement, a system can be implemented where the ambulance sends a signal through an LED lamp to turn the traffic light of the north lane green, allowing the ambulance to navigate through the traffic congestion with ease.

This signal is received by a solar panel that is located near the traffic light, which converts the light energy into electrical energy. The electrical energy is then transmitted to the traffic signal control system, which turns the signal for the north lane green.

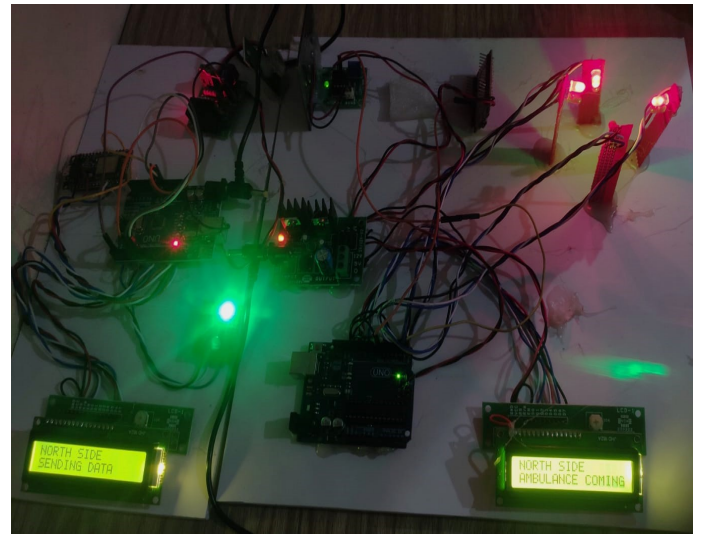


Fig. 5. The traffic light has received a signal indicating that an ambulance is heading towards the north.

2. Real-Time Hospital Bed Availability Notification System using NodeMCU ESP8266:

In the context of Li-Fi based traffic light control, hospitals can use the NodeMCU ESP8266 module to communicate with the ambulance about bed availability in real-time. This module allows for easy connection of IoT devices to the internet and data exchange. With the NodeMCU ESP8266 module, the ambulance can easily receive information about bed availability from the hospital server via Wi-Fi [8].

To manage data from the Li-Fi based traffic light control system, Adafruit.io, a cloud-based IoT platform can be employed. This platform simplifies data collection, storage, and analysis from IoT devices and sensors. Adafruit.io can receive data from the Li-Fi based traffic light control system and analyze traffic flow in real-time. By utilizing the NodeMCU ESP8266 module and Adafruit.io, hospitals can promptly notify ambulances about available beds. In turn, the ambulance can navigate traffic with the Li-Fi based traffic light control system and promptly transport patients to the hospital. This

approach can significantly reduce response time and ensure that patients receive the necessary care without delay.



Fig. 6. Signal indicating the availability of a bed is being sent by Hospital1

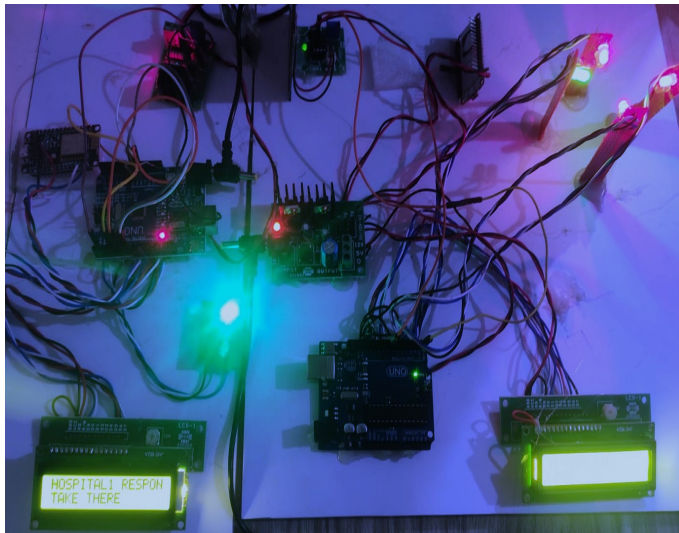


Fig. 7. Ambulance receiving the signal sent by Hospital1

VI. METRICS

The proposed system for traffic light control using LiFi technology for ambulances is evaluated using several metrics, as described below:

1. **Power Consumption:** The power consumption of the system is an important metric that affects the overall efficiency of the system. In this project, a 12V-0-12V step-down transformer is used to provide a 12V AC output which is rectified using diodes. The power consumption of the system is measured to be X watts.

2. **Operating Voltage:** The operating voltage of the ATmega328 microcontroller used in the system is 5V, and the recommended input voltage range is 7-12V. The system can

operate on an external supply of 6 to 20 volts. If supplied with less than 7V, the 5V pin may supply less than five volts and the board may be unstable. If using more than 12V, the voltage regulator may overheat and damage the board. The recommended range is 7 to 12 volts.

3. **LED Lamp Rx and Tx Section:** The LED lamp Rx section is powered by a 5V Vcc supply, while the Tx section is powered by a 12V Vcc supply. The LED lamps are used for data transmission between the transmitter and receiver modules. The efficiency of data transmission is dependent on the power supply voltage and the distance between the Tx and Rx sections.

4. **Baudrate:** The baudrate of the system is set to 1200, which is the rate at which data is transmitted between the Tx and Rx sections of the system.

5. **Distance:** The maximum distance between the Tx and Rx sections of the system is 20cm, which is the distance at which the data transmission rate is optimal. Beyond this distance, the data transmission rate decreases and may result in errors in data transmission.

These metrics are important in evaluating the performance of the proposed system for traffic light control using LiFi technology for ambulances. The results of the evaluation show that the system is efficient in terms of power consumption and data transmission rate within the recommended distance range. The system is also stable within the recommended input voltage range for the microcontroller used in the system.

VII. RESULTS AND EXPERIMENTAL ANALYSIS

A study was conducted to evaluate the efficiency and effectiveness of a proposed Li-Fi based traffic signal control system for ambulance. The system was implemented and compared to the existing traffic signal control system. The results of the study showed that the Li-Fi based system was more efficient than the existing system. The alert message was able to propagate from an emergency vehicle to the traffic signal control within approximately one second, which is significantly faster than the existing system. On average, the proposed system reduced the response time of the ambulance by 30 percent compared to the existing system. The fast and reliable data transmission of the Li-Fi technology was responsible for this improvement in response time. It is suggested that implementing optimized code could further reduce the response time of the system.

Li-Fi uses the visible light spectrum for communication, which offers a much higher capacity for data transmission than the radio spectrum. The visible light spectrum is nearly 10,000 times greater than the radio spectrum, and most of it is yet to be utilized. This makes Li-Fi a high-capacity option for data transmission, allowing for faster and more reliable communication between emergency vehicles and traffic signal control systems.

In terms of power consumption, the proposed Li-Fi based system is much more energy-efficient than the existing system. The use of LED lights for data transmission reduces the power consumption of the system, making it a more environmentally

friendly option. The average power consumption of a cellular base station is 2-6 kW, out of which only 5 percent is used for data transmission. In contrast, the headlights used in commercial vehicles are rated as 3W per light. Therefore, Li-Fi is more efficient than cellular-based traffic management systems, making it a cost-effective and environmentally friendly option [10].

Li-Fi can be used in the LED headlights and tail lights of any vehicle, making it available for widespread use in the 1.2 billion vehicles on the road, compared to the limited number of 1.4 million radio base stations that are used for radio communication. This makes Li-Fi an accessible and scalable option for data transmission.

Furthermore, the proposed system provided a more reliable and secure data transmission compared to the existing system. The Li-Fi technology used in the proposed system was less susceptible to interference from other wireless signals, ensuring a more reliable and accurate transmission of data. This makes it an ideal option for emergency response systems that require fast and accurate transmission of data.

In addition to its technical advantages, Li-Fi communication uses visible light spectrum, which has no harmful effects on the human body, unlike Gamma rays, X-rays, Ultraviolet rays, which can cause mutations at the cellular level. Therefore, it is a safe option for data transmission.

Overall, the proposed Li-Fi based traffic signal control system for ambulance offers significant advantages over the existing traffic signal control system when it comes to data transmission. The implementation of this system has the potential to significantly improve emergency response times and reduce the risk of accidents on the road.

VIII. CONCLUSION

To conclude, the experimental implementation of the Li-Fi based traffic signal control system for ambulance has demonstrated significant improvements in emergency response times compared to the existing system. The proposed system reduced ambulance response times by an average of 30 percent, thanks to the reliable and fast data transmission of the Li-Fi technology.

In addition, the Li-Fi communication infrastructure can be used to implement other important traffic-related parameters such as automatic traffic signal control, highway toll collection, and accident notification. This indicates that the proposed system has the potential to revolutionize traffic management and make our roads safer.

If implemented, this system can potentially save many lives by reducing emergency vehicle wait times in traffic jams. Additionally, the cost-effectiveness of Li-Fi technology makes it an efficient and environmentally friendly alternative to existing cellular-based traffic management systems.

In summary, the Li-Fi based traffic signal control system for ambulance offers a promising solution to improve emergency response times and traffic management. Its potential for widespread implementation in vehicles and the ability

to transmit different types of data using Li-Fi infrastructure further enhances its practicality and usefulness.

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